In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_openml
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
```

In [6]:
```
boston=fetch_openml(
    name="boston",
    version=1,
    as_frame=False
)
X=boston.data
y=boston.target.astype(float)
```

In [7]:
```
X=X[:,5].reshape(-1,1)
```

In [8]:
```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state
```

In [9]:
```
degrees=[1,2,3,4,5,8,10]
train_errors=[]
val_errors=[]
for degree in degrees:
    poly=PolynomialFeatures(degree=degree)
    X_train_poly=poly.fit_transform(X_train)
    X_val_poly=poly.transform(X_test)
    model=LinearRegression()
    model.fit(X_train_poly,y_train)
    train_pred=model.predict(X_train_poly)
    val_pred=model.predict(X_val_poly)
    train_error=mean_squared_error(y_train,train_pred)
    val_error=mean_squared_error(y_test,val_pred)
    train_errors.append(train_error)
    val_errors.append(val_error)

    print(
        "Degree:",degree,
        "Training Error:",round(train_error,2),
        "Validation Error:",round(val_error,2)
    )
```

```
Degree: 1 Training Error: 43.0 Validation Error: 46.14
Degree: 2 Training Error: 38.91 Validation Error: 35.37
Degree: 3 Training Error: 37.73 Validation Error: 34.45
Degree: 4 Training Error: 35.89 Validation Error: 39.58
Degree: 5 Training Error: 35.15 Validation Error: 32.51
Degree: 8 Training Error: 34.97 Validation Error: 34.7
Degree: 10 Training Error: 34.26 Validation Error: 152.29
```

In [10]:
```
plt.figure(figsize=(8,5))
plt.plot(degrees,train_errors,marker='o',label="Training Error")
plt.plot(degrees,val_errors,marker='o',label="Validation Error")
```

```
plt.xlabel("Degrees")
plt.ylabel("Error")
plt.title("Bias-Variance Tradeoff")
plt.legend()
plt.grid()
plt.show()
```



In [ ]: